

Absolute Convergence; The Ratio and Root tests:

When some of the terms of a series are positive and others are negative, the series may or may not converge.

for example, the series

$$5 - \frac{5}{4} + \frac{5}{16} - \frac{5}{64} + \dots = \sum_{n=0}^{\infty} 5 \left(-\frac{1}{4}\right)^n \rightarrow \textcircled{1}$$

converges ($\because |r| < 1$), whereas the series

$$1 - \frac{5}{4} + \frac{25}{16} - \frac{125}{64} + \dots = \sum_{n=0}^{\infty} \left(-\frac{5}{4}\right)^n \rightarrow \textcircled{2}$$

diverges ($\because |r| > 1$)

In $\textcircled{1}$, there is ~~some~~ cancellation in the partial sums, which may be affecting the convergence property of the series.

However, if we make all of the terms positive in series $\textcircled{1}$ to form the new series

$$5 + \frac{5}{4} + \frac{5}{16} + \frac{5}{64} + \dots = \sum_{n=0}^{\infty} \left| 5 \left(-\frac{1}{4}\right)^n \right| = \sum_{n=0}^{\infty} 5 \left(\frac{1}{4}\right)^n$$

we see that it still converges.

Defⁿ:- A series $\sum a_n$ converges absolutely (is absolutely convergent) if the corresponding series of absolute values

$\sum |a_n|$, converges.

So the geometric series ① is absolutely convergent. We observed, it is also convergent.

Theorem:- The Absolute Convergence Test

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

Proof:- For each n , $-|a_n| \leq a_n \leq |a_n|$

$$\Rightarrow 0 \leq a_n + |a_n| \leq 2|a_n|$$

Since $\sum_{n=1}^{\infty} |a_n|$ converges and therefore by direct comparison test $\sum_{n=1}^{\infty} (a_n + |a_n|)$ converges.

we write $a_n = (a_n + |a_n|) - |a_n|$

$$\Rightarrow \sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (a_n + |a_n|) - \sum_{n=1}^{\infty} |a_n|$$

Since the difference of two convergent series is convergent and therefore $\sum_{n=1}^{\infty} a_n$ converges.

Example Test the absolute convergence of the series

i) $\sum_{n=1}^{\infty} (-1)^{n+1} \cdot \frac{1}{n^2} = 1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \dots$

ii) $\sum_{n=1}^{\infty} \frac{\sin n}{n^2} = \frac{\sin 1}{1^2} + \frac{\sin 2}{2^2} + \dots$

Sol:- i), the corresponding series of absolute values sum

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{1} + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots \text{ convergent by}$$

p-series test and therefore the original series converges absolutely.

ii) The corresponding series of absolute values is

$$\left| \frac{\sin 1}{1^2} \right| + \left| \frac{\sin 2}{2^2} \right| + \left| \frac{\sin 3}{3^2} \right| + \dots = \sum_{n=1}^{\infty} \left| \frac{\sin n}{n^2} \right|$$

$$\text{Now } \sum_{n=1}^{\infty} \left| \frac{\sin n}{n^2} \right| \leq \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ because } |\sin n| \leq 1 \quad \forall n$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ convergent series and therefore $\sum_{n=1}^{\infty} \left| \frac{\sin n}{n^2} \right|$

convergent by Direct Comparison Test. Therefore the original series $\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$ converges absolutely.

Example:- Test the ^{absolute} convergence of the series:

i) $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

ii) $\frac{1}{1 \cdot 3} - \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 5} - \frac{1}{4 \cdot 6} + \dots$

iii) $\sum_{n=1}^{\infty} (-1)^{n-1} \cdot \frac{1}{n \sqrt{n}}$

Sol:- i), the corresponding ^{series of} absolute values is

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots = \sum_{n=1}^{\infty} \frac{1}{n}$$

which is divergent by p-series test. Therefore, the original series not absolutely convergent.

(ii) The corresponding series of absolute values is

$$\frac{1}{1 \cdot 3} + \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 5} + \frac{1}{4 \cdot 6} + \dots = \sum_{n=1}^{\infty} \frac{1}{n(n+2)}$$

$$\text{Since } \frac{1}{n(n+2)} < \frac{1}{n^2} \quad \forall n$$

By Direct-Comparison Test $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$ converges because $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges by p-series test.

$\therefore$ The original series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n(n+2)}$ converges absolutely.

$$(iii) \sum_{n=1}^{\infty} \left| (-1)^{n+1} \frac{1}{n\sqrt{n}} \right| = \sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}}$$

$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ converges by p-series Test.

$\therefore$ the original series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n(n+2)}$ converges absolutely.

EX: Test the absolute convergence of the following series:

$$i) \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{n^2+1} \quad ii) \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n!}$$

The Ratio Test:-

The Ratio Test measures the rate of growth (or decline) of a series by examining the ratio $\frac{a_{n+1}}{a_n}$.

For a geometric series $\sum ar^n$, this rate is constant ($\frac{a_{n+1}}{a_n} = r$), and the series converges $\Leftrightarrow$ its ratio is less than 1 in absolute value.

Theorem:- The Ratio Test

Let $\sum a_n$ be any series and suppose that $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = p$.

Then (a) the series converges absolutely if $p < 1$,

(b) the series diverges if $p > 1$ or p is infinite,

(c) the test is inconclusive if $p = 1$.

Note:- When $p = 1$, the test is inconclusive. In this case, some other test must be used to determine convergence of the series.

* The Ratio Test is often effective when the terms of a series contain factorials or expressions involving n or expressions raised to a power involving n .

Example:- Test the convergence of the following series:

$$i) \sum_{n=1}^{\infty} \frac{1}{n}$$

$$ii) \sum_{n=1}^{\infty} \frac{1}{n^2}$$

Sol: i) $a_n = \frac{1}{n}$, then $a_{n+1} = \frac{1}{n+1}$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \quad (p=1)$$

The Ratio Test is inconclusive, but the series $\sum_{n=1}^{\infty} \frac{1}{n}$ divergent by p-series test.

ii, please try yourself.

Example: Investigate the convergence of the following series:

$$i) \sum_{n=0}^{\infty} \frac{2^n + 5}{3^n}$$

$$ii) \sum_{n=1}^{\infty} \frac{2^n}{n!}$$

$$iii) \sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2}$$

$$iv) \sum_{n=1}^{\infty} \frac{n!}{n^n}$$

Sol: i) $a_n = \frac{2^n + 5}{3^n}$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{2^{n+1} + 5}{3^{n+1}} \times \frac{3^n}{2^n + 5} \right| = \frac{1}{3} \left(\frac{2^{n+1} + 5}{2^n + 5} \right)$$

$$= \frac{1}{3} \left(\frac{2 + \frac{5}{2^n}}{1 + \frac{5}{3^n}} \right)$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{1}{3} \left(\frac{2 + \frac{5}{2^n}}{1 + \frac{5}{2^n}} \right) = \frac{1}{3} \cdot 2 = \frac{2}{3} < 1$$

$\therefore$ the series $\sum_{n=1}^{\infty} \frac{2^n + 5}{3^n}$ converges absolutely.

ii) Here $a_n = \frac{2^n}{n!}$

$$\text{Now } \left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{2^{n+1}/(n+1)!}{2^n/n!} \right| = \frac{2}{n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{2}{n+1} = 0 < 1$$

$\therefore \sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges absolutely by Ratio Test.

iii) Here $a_n = \frac{(2n)!}{(n!)^2}$, $a_{n+1} = \frac{(2(n+1))!}{((n+1)!)^2}$

$$\begin{aligned} \left| \frac{a_{n+1}}{a_n} \right| &= \frac{(2n+2)(2n+1)(2n)!}{(n+1)^2 \cdot (n!)^2} \times \frac{(n!)^2}{(2n)!} \\ &= \frac{(2n+2)(2n+1)}{(n+1)^2} \end{aligned}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{4n^2 + 6n + 2}{n^2 + 2n + 1}$$

$$= 4 > 1$$

$\therefore$ the series $\sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2}$ diverges by Ratio Test.

(iv), please try yourself.

Exercise: Discuss the convergence of the following series:

i) $\frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2^{n-1}+1} + \dots$

ii) $\frac{1^2 \cdot 2^2}{1!} + \frac{2^2 \cdot 3^2}{2!} + \frac{3^2 \cdot 4^2}{3!} + \frac{4^2 \cdot 5^2}{4!} + \dots$

iii) $\sum_{n=1}^{\infty} \frac{1}{n!}$

iv) $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$

v) $\sum_{n=1}^{\infty} \frac{5^n}{n^2+7}$

Example: Test the convergence of the series $\sum_{n=1}^{\infty} a_n$, where

$$a_n = \begin{cases} \frac{n}{2^n}, & n \text{ odd} \\ \frac{1}{2^n}, & n \text{ even} \end{cases}$$

Sol: To investigate convergence we write out several terms of the series

$$\sum_{n=1}^{\infty} a_n = \frac{1}{2^1} + \frac{1}{2^2} + \frac{3}{2^3} + \frac{1}{2^4} + \frac{5}{2^5} + \frac{1}{2^6} + \dots$$

$$= \frac{1}{2} + \frac{1}{4} + \frac{3}{8} + \frac{1}{16} + \frac{5}{32} + \frac{1}{64} + \dots$$

clearly, this is not a geometric series. The n th term ~~is~~ approaches zero as $n \rightarrow \infty$ so the n th term Test

does not work. The Integral Test does not look promising. Therefore, we will try to apply Ratio Test.

The Ratio Test produces $\left| \frac{a_{n+1}}{a_n} \right| = \begin{cases} \frac{1}{2^n}, & n \text{ odd} \\ \frac{n+1}{2}, & n \text{ even} \end{cases}$

When n is odd (so $n+1$ is even)

$$a_n = \frac{n}{2^n}, \quad a_{n+1} = \frac{1}{2^{n+1}}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{1}{2^{n+1}} \times \frac{2^n}{n} \right| = \frac{1}{2n}$$

When n is even (so $n+1$ is odd)

$$a_n = \frac{1}{2^n}, \quad a_{n+1} = \frac{n+1}{2^{n+1}}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{n+1}{2^{n+1}} \times \frac{2^n}{1} \right| = \frac{n+1}{2}$$

As $n \rightarrow \infty$, the ratios are alternatively small and large and therefore has no limit. Hence Ratio Test fails to conclude. We will see that the following test establishes that the series converges.

Theorem:- The Root Test

Let $\sum a_n$ be any series and suppose that

$$\lim_{n \rightarrow \infty} (|a_n|)^{1/n} = \rho.$$

then (a) the series converges absolutely if $p < 1$,

(b) the series diverges if $p > 1$ or p is infinite

(c) the test is inconclusive if $p = 1$.

Example:- Consider again the series with terms

$$a_n = \begin{cases} \frac{n}{2^n}, & n \text{ odd} \\ \frac{1}{2^n}, & n \text{ even} \end{cases} \quad \text{Does } \sum a_n \text{ converge?}$$

Sol:- We will apply the Root Test,

$$|a_n|^{1/n} = \begin{cases} \frac{n^{1/n}}{2}, & n \text{ odd} \\ \frac{1}{2}, & n \text{ even} \end{cases}$$

$$\therefore \frac{1}{2} \leq |a_n|^{1/n} \leq \frac{n^{1/n}}{2}$$

Since $n^{1/n} \rightarrow 1$ as $n \rightarrow \infty$. Therefore, by the Sandwich

theorem $\lim_{n \rightarrow \infty} |a_n|^{1/n} = \frac{1}{2} < 1$, so the series converges

absolutely by the ~~Root~~ Root Test.

Example:- Test the convergence of the following series:

i) $\sum \left(\frac{n}{n+1} \right)^{n^2}$

ii) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

$$(iii) \sum_{n=1}^{\infty} \frac{2^n}{n^3}$$

$$(iv) \sum_{n=1}^{\infty} \left(\frac{1}{1+n} \right)^n$$

$$(v) \sum_{n=2}^{\infty} \frac{1}{(\ln n)^n}$$

$$(vi) \sum \left(\frac{n+1}{3n} \right)^n$$

$$(vii) \sum \left(1 + \frac{1}{\sqrt{n}} \right)^{-n^{3/2}}$$

Sol: i) Here $a_n = \left(\frac{n}{n+1} \right)^{n^2}$

$$\lim_{n \rightarrow \infty} (|a_n|)^{1/n} = \lim_{n \rightarrow \infty} \left[\left(\frac{n}{n+1} \right)^{n^2} \right]^{1/n}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = \lim_{n \rightarrow \infty} \left(\frac{1}{1 + \frac{1}{n}} \right)^n$$

$$= \frac{1}{e} < 1$$

$\therefore$ By Cauchy's Root Test, the given series $\sum a_n$ is convergent (or absolutely convergent)

ii) Here $a_n = \frac{n^2}{2^n}$

$$\lim_{n \rightarrow \infty} (|a_n|)^{1/n} = \lim_{n \rightarrow \infty} \left(\frac{n^2}{2^n} \right)^{1/n} = \lim_{n \rightarrow \infty} \frac{n^{2/n}}{2}$$

$$= \frac{1}{2} \lim_{n \rightarrow \infty} n^{2/n}$$

$$= \frac{1}{2} < 1$$

$\therefore$ By Root Test, the series $\sum a_n$ converges absolutely.

(iii) please try yourself.

(iv) please try yourself.

(v) Here $a_n = \frac{1}{(\ln n)^n}$

$$\lim_{n \rightarrow \infty} (|a_n|)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left[\frac{1}{(\ln n)^n} \right]^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{1}{\ln n} = 0 < 1$$

By Root Test, the series $\sum \frac{1}{(\ln n)^n}$ is absolutely convergent.

(vi) please try yourself.

(vii) please try yourself.

Example:- Test for absolute convergence of the following series:

i) $\sum 5^{-n} - (-1)^n$

ii) $\sum_{n=1}^{\infty} 3^{2n} - 5(-1)^n$

Sol:- i) Here $a_n = 5^{-n} - (-1)^n$

$$|a_n|^{\frac{1}{n}} = |5^{-n} - (-1)^n|^{\frac{1}{n}} = 5^{1 - \frac{(-1)^n}{n}}$$

$$\begin{aligned}\lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} &= \lim_{n \rightarrow \infty} 5^{1 - \frac{(-1)^n}{n}} \\ &= 5^{1-0} = \frac{1}{5} < 1\end{aligned}$$

$\therefore$ By Root Test, the series is absolutely convergent.

iii) please try yourself.

Example: show that Root Test establishes the convergence of the series $\sum a_n$, where

$$a_n = \begin{cases} \frac{1}{2}^n, & n \text{ is odd} \\ \frac{1}{2}^{n+2} & n \text{ is even} \end{cases}$$

while Ratio Test fails to do so.

Sol: When n is odd, $\lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(\frac{1}{2}\right)^{\frac{1}{n}}$

$$= \lim_{n \rightarrow \infty} \frac{1}{2} = \frac{1}{2} < 1$$

When n is even, $\lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(\frac{1}{2}^{n+2}\right)^{\frac{1}{n}}$

$$= \lim_{n \rightarrow \infty} \frac{1}{2} \cdot 2^{\frac{2}{n}} \text{ (or) } 2^{-1 + \frac{2}{n}}$$

$$= \frac{1}{2}^{-1+0} = \frac{1}{2} < 1$$

$$\therefore \lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \frac{1}{2} < 1 \quad \forall n, \text{ even or odd}$$

$\therefore$ By Root Test, $\sum a_n$ is absolutely convergent.

When n is odd (so that $n+1$ is even)

$$a_n = 2^{-n}, \quad a_{n+1} = 2^{-(n+1)+2} = 2^{-n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{2^{-n+1}}{2^{-n}} = 2 > 1$$

When n is even (so that $n+1$ is odd)

$$a_n = 2^{-n+2} \text{ and } a_{n+1} = 2^{-(n+1)} = 2^{-n-1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{2^{-n-1}}{2^{-n+2}} \right| = \frac{1}{2^3} = \frac{1}{8} < 1$$

Thus $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ does not exist since we have different results for n odd and even.

$\therefore$ Ratio Test fails.

Example:- Show that Root Test establishes the convergence of the series $\sum 3^n - (-1)^n$ while Ratio Test fails to do so.

Sol:- please try yourself.

(45)

Note:- Root Test is more general than Ratio Test (or)
Root Test is stronger than Ratio Test.
This is so because

$$i, \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \text{ exists} \Rightarrow \lim_{n \rightarrow \infty} (|a_n|)^{\frac{1}{n}} \text{ exists}$$

$$\text{and } \lim_{n \rightarrow \infty} (|a_n|)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \quad (\because \text{Cauchy's second theorem on limits})$$

Whenever Ratio Test is applicable, so is the Root Test.

$$ii, \lim_{n \rightarrow \infty} (|a_n|)^{\frac{1}{n}} \text{ exists need not imply } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \text{ exists.}$$

$$\text{i.e., } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \text{ may not exist.}$$

When the Ratio Test fails, the Root Test ~~succeeds~~.

Hence the Root Test is more general than Ratio Test

Alternating Series and Conditional Convergence

A series in which the terms are alternatively positive and negative is an alternating series.

The n th term of an alternating series $\sum a_n$ is of the form

$$a_n = (-1)^{n+1} u_n \quad \text{or} \quad a_n = (-1)^n u_n$$

where $u_n = |a_n|$ is a positive number.

Some Examples for alternating series:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots + (-1)^{n+1} \frac{1}{n} + \dots \rightarrow \textcircled{1}$$

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \dots + (-1)^{n+1} \frac{1}{2^{n-1}} + \dots \rightarrow \textcircled{2}$$

$$1 - 2 + 3 - 4 + 5 - 6 + \dots + (-1)^{n+1} n + \dots \rightarrow \textcircled{3}$$

series $\textcircled{2}$, a geometric series with ratio $r = -\frac{1}{2}$ converges to $\frac{1}{1 - (-\frac{1}{2})} = \frac{2}{3}$. Series $\textcircled{3}$ diverges because the n th

term does not approach zero. Series $\textcircled{1}$, called the alternating harmonic series. Alternating harmonic series is convergent and this we will prove by applying alternating series test. This test is for convergence of an alternating series and cannot be used to conclude that such a series diverges.

Theorem: The Alternating Series Test

The series $\sum_{n=1}^{\infty} (-1)^{n+1} u_n = u_1 - u_2 + u_3 - u_4 + \dots$

converges if the following conditions are satisfied:

i) The u_n 's are all positive.

ii) The u_n 's are eventually nonincreasing: $u_n \geq u_{n+1}$ for all $n \geq N$, for some integer N .

iii) $u_n \rightarrow 0$ i.e., $\lim_{n \rightarrow \infty} u_n = 0$

proof:- The series $\sum_{n=1}^{\infty} (-1)^{n+1} u_n = u_1 - u_2 + u_3 - u_4 + \dots$

where $u_1, u_2, u_3, u_4, \dots$ is nonincreasing, so that $N=1$.

The n th partial sum of the series $\sum_{n=1}^{\infty} (-1)^{n+1} u_n$ is

$$S_n = u_1 - u_2 + u_3 - u_4 + \dots + (-1)^n u_n$$

If n is an even integer, say $n=2m$, then the first n terms summation is

$$S_{2m} = (u_1 - u_2) + (u_3 - u_4) + \dots + (u_{2m-1} - u_{2m})$$

$$= u_1 - (u_2 - u_3) - (u_4 - u_5) - \dots - (u_{2m-2} - u_{2m-1}) - u_{2m}$$

The first equality shows that S_{2m} is the sum of m nonnegative terms, since each term in the parentheses is positive or zero. Hence $S_{2m+2} \geq S_{2m}$ and the sequence $\{S_{2m}\}$ is nondecreasing.

The second Equality shows that $s_{2m} \leq u$, and therefore the sequence $\{s_{2m}\}$ is bounded from above.

Since $\{s_{2m}\}$ is nondecreasing and bounded from above, therefore, it is convergent and say it converges to L

$$\text{i.e., } \lim_{m \rightarrow \infty} s_{2m} = L \rightarrow \textcircled{a}$$

If n is odd integer, say $n = 2m+1$, then the sum of the first n terms is $s_{2m+1} = s_{2m} + u_{2m+1}$.

Since $u_n \rightarrow 0$ (condition iii) of the theorem)

$$\therefore \lim_{m \rightarrow \infty} u_{2m+1} = 0 \text{ and}$$

$$\text{as } m \rightarrow \infty \quad s_{2m+1} = s_{2m} + u_{2m+1} \rightarrow L + 0 = L$$

$$\therefore \lim_{m \rightarrow \infty} s_{2m+1} = L \rightarrow \textcircled{b}$$

Combining the results of $\textcircled{a}$ and $\textcircled{b}$ gives $\lim_{n \rightarrow \infty} s_n = L$.

$\Rightarrow$ ~~Therefore~~, the sequence of partial sums $\{s_n\}$ converges.

Hence, $\sum_{n=1}^{\infty} (-1)^{n+1} u_n$ converges.

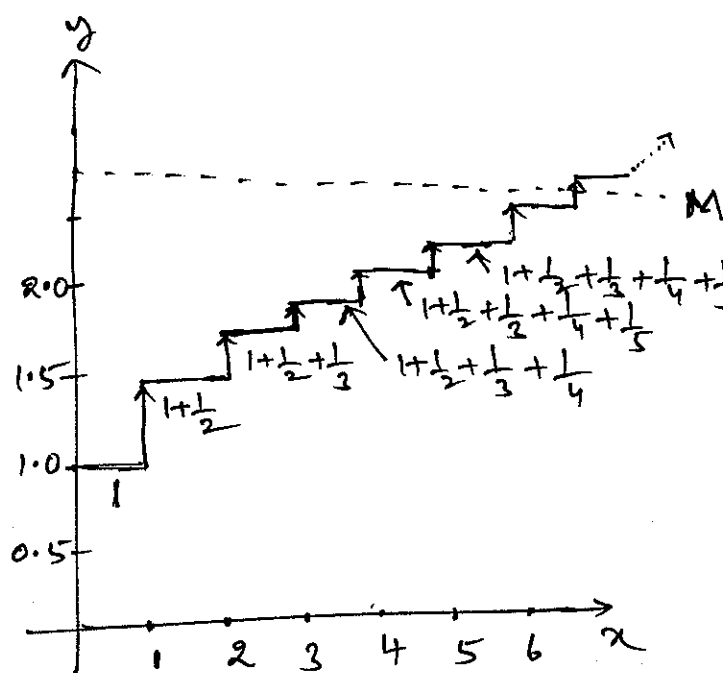
Important Note:- The Alternating Series Test gives no information about what the sum of the series might be.

Example:- The alternating harmonic series

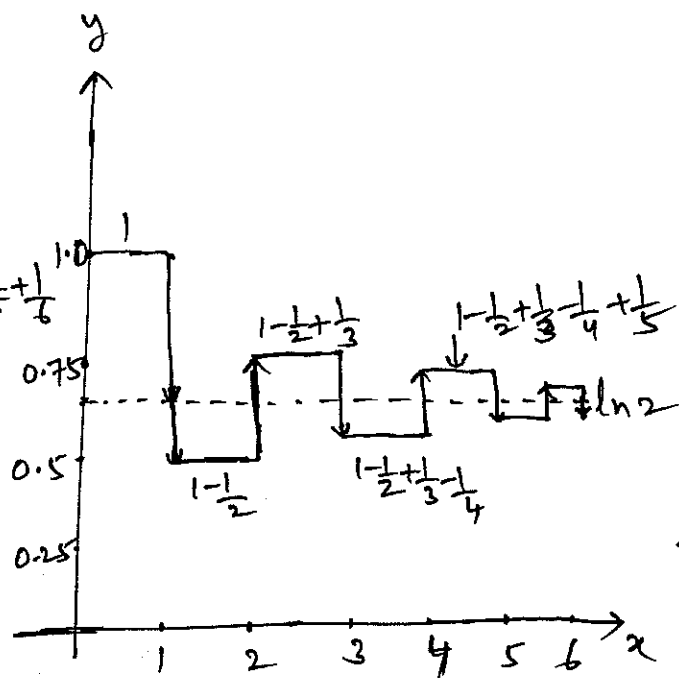
$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \text{ clearly}$$

satisfies the three requirements of Alternating Series Test with $N=1$; it therefore converges. But no information about the sum of the series.

* Histograms of the partial sums of the divergent harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ and convergent alternating harmonic series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$



(a)



(b)

In Fig. (a), the harmonic series diverges, with partial sums that eventually exceed any constant.

In Fig. (b), The alternating harmonic series converges to $\ln 2 \approx 0.693$

Example 1:- Test the convergence of an alternating series:

i) $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$

ii) $\frac{1}{1 \cdot 3} - \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 5} - \frac{1}{4 \cdot 6} + \dots$

iii) $\sum_{n=1}^{\infty} (-1)^{n+1} \sin\left(\frac{1}{n}\right)$

Sol:- i) The given series is $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{2n-1} = \sum_{n=1}^{\infty} (-1)^{n+1} u_n$

Here $u_n = \frac{1}{2n-1}$

clearly, $u_n = \frac{1}{2n-1} > 0 \forall n$

and $u_{n+1} = \frac{1}{2(n+1)-1} = \frac{1}{2n+1}$

Since $2n+1 > 2n-1 \forall n$

$$\Rightarrow \frac{1}{2n-1} > \frac{1}{2n+1} \forall n$$

$$\Rightarrow u_n > u_{n+1} \forall n$$

Also, $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{2n-1} = 0$

$\therefore$ All the requirements of the Alternating Series Test are satisfied.

Hence the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{2n-1}$ is convergent by Alternating Series Test.

Sol-ii, The given series is $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n(n+2)}$

Here $u_n = \frac{1}{n(n+2)} > 0 \forall n$

$$u_{n+1} = \frac{1}{(n+1)(n+3)}$$

Since $\frac{1}{n} > \frac{1}{n+1}$ and $\frac{1}{n+2} > \frac{1}{n+3}$

$$\therefore \frac{1}{n(n+2)} > \frac{1}{(n+1)(n+3)}$$

$$\Rightarrow u_n > u_{n+1} \forall n.$$

Also, $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{n(n+2)} = 0$

$\therefore$ By Alternating Series Test, the given series

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n(n+2)} = \frac{1}{1 \cdot 3} - \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 5} - \frac{1}{4 \cdot 6} + \dots \text{ is convergent.}$$

Solⁱⁱⁱ please try yourself.

* $\rightarrow$ We know that the second condition of Alternating Series Test, i.e., $u_n \geq u_{n+1}, \forall n \geq N$ for some integer N .
Sometimes rather than directly verifying this condition $u_n \geq u_{n+1}$, a second way to show that the sequence $\{u_n\}$ is nonincreasing is to define a differentiable function $f(x)$ satisfying $f(n) = u_n$. If $f'(x) \leq 0$ for all x greater than or equal to some positive integer N , then $f(x)$ is nonincreasing for $x \geq N$. It follows that $f(n) \geq f(n+1)$, or $u_n \geq u_{n+1}$, for $n \geq N$.

Example: Test the convergence of $\sum_{n=1}^{\infty} (-1)^{n+1} \cdot \frac{n}{n^2+16}$

Sol: Here $u_n = \frac{n}{n^2+16}$ by comparing with $\sum_{n=1}^{\infty} (-1)^{n+1} u_n$.

$$u_n = \frac{n}{n^2+16} > 0 \quad \forall n$$

$$\text{Let } u_n = f(n) = \frac{n}{n^2+16}$$

$$\text{define } f(x) = \frac{x}{x^2+16}$$

$$f'(x) = \frac{16-x^2}{(x^2+16)^2} \leq 0 \quad \forall x \geq 4$$

$\therefore f(x)$ is nonincreasing for $x > 4$

It follows that $u_n \geq u_{n+1} \forall n > 4$

Also, $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{n}{n^2 + 16} = \lim_{n \rightarrow \infty} \frac{1}{n \left[1 + \frac{16}{n^2} \right]} = 0.$

$\therefore$ By ~~Alternating~~ Series Test, the given series converges.

Graphical Representation

Consider an alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} u_n$ whose partial sums are

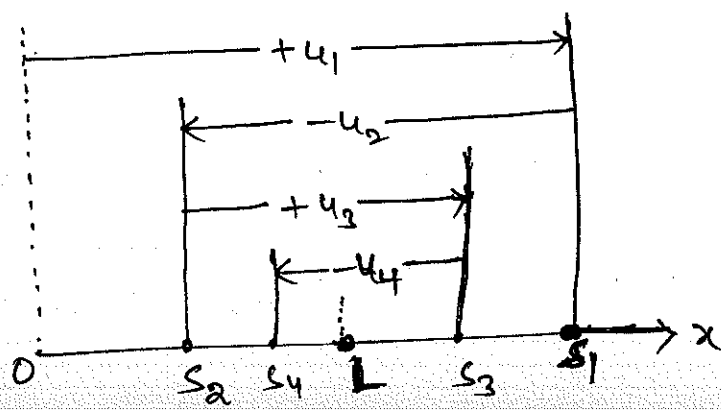
$$S_1 = u_1$$

$$S_2 = u_1 - u_2$$

$$S_3 = u_1 - u_2 + u_3$$

$\vdots$

$$S_n = u_1 - u_2 + u_3 - u_4 + \dots + (-1)^{n+1} u_n$$



A graphical interpretation of the partial sums (Figure) shows how an alternating series converges to its limit L when the three conditions of the Alternating series Test are satisfied.

Starting from the origin of the x -axis, we lay off the positive distance $s_1 = u_1$. To find the point corresponding to $s_2 = u_2 - u_1$, we back up a distance equal to u_2 . Since $u_2 \leq u_1$, we do not back up any farther than the origin.

We continue in this seesaw fashion, backing up or going forward as the signs in the series demand. But for $n \geq N$, each forward or backward step is shorter than the preceding step because $u_{n+1} \leq u_n$. And since the n th term approaches zero as n increases, the size of the step we take forward or backward gets smaller and smaller. We oscillate back and forth across the limit L , and the amplitude of oscillation approaches zero. The limit L lies between any two successive sums s_n and s_{n+1} and hence differs from s_n by an amount less than u_{n+1} , i.e. $|L - s_n| < u_{n+1}$ for $n \geq N$.

Theorem:- The alternating series estimation theorem (50)

If the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} u_n$ satisfies the three conditions of Alternating Series Test, then for $n \geq N$,

$$S_n = u_1 - u_2 + u_3 - u_4 + \dots + (-1)^{n+1} u_n$$

approximates the sum L of the series with an error whose absolute value is less than u_{n+1} , the absolute value of the first unused term. Furthermore, the sum L lies between any two successive partial sums S_n and S_{n+1} , and the remainder, $L - S_n$ has the same sign as the first unused term.

Example:- We apply above estimation theorem on a series

$$\sum_{n=0}^{\infty} (-1)^n \frac{1}{2^n} = 1 - \frac{1}{2} + \frac{1}{2^2} - \frac{1}{2^3} + \frac{1}{2^4} - \frac{1}{2^5} + \frac{1}{2^6} - \frac{1}{2^7} + \frac{1}{2^8} - \dots$$

$$= 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} + \frac{1}{64} - \frac{1}{128} + \frac{1}{256} - \dots$$

If we truncate the series after the eighth term, we throw away a total that is positive and less than $\frac{1}{256}$. The sum of the first eight terms is $S_8 = 0.6640625$ and

the sum of the first nine terms $\epsilon/s \quad S_9 = 0.66796875$.

The sum of the geometric series $\epsilon/s \quad \frac{1}{1 - (-\frac{1}{2})} = \frac{2}{3}$, and

we note that $0.6640625 < \frac{2}{3} < 0.66796875$.

the difference of $\frac{2}{3} - 0.6640625 = 0.0026041666\dots$,

ϵ/s positive and less than $\frac{1}{256} = 0.00390625$.

Conditional Convergence:-

If we replace all the negative terms in the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{2^{n-1}}$, changing them to positive terms

instead, we obtain the geometric series $\sum_{n=1}^{\infty} \frac{1}{2^n}$. The

original and new series of absolute values both converge (although to different sums). For an absolutely

convergent series, changing infinitely many of the negative terms in the series to positive values does not change its property of still being a convergent series. Other convergent series may behave differently.

The convergent alternating harmonic series has infinitely many negative terms, but if we change its negative terms to positive values, the resulting series is the divergent harmonic series. So the presence of infinitely many negative terms is essential to the